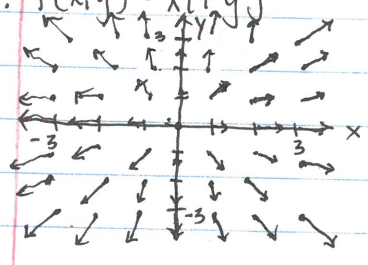
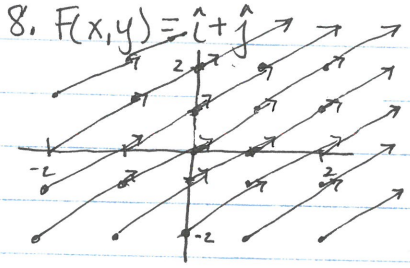
WP 14 Solutions

5. $F(x,y) = x\hat{i} + y\hat{j}$



8. $F(x,y) = \hat{i} + \hat{j}$



23. All vectors point toward origin and have constant length

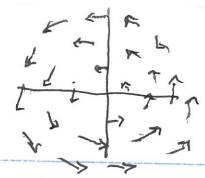
$$F(x,y) = \frac{y}{\sqrt{x^2+y^2}}\hat{i} - \frac{x}{\sqrt{x^2+y^2}}\hat{j}$$

25. $F(x,y) = M(x,y)\hat{i} + N(x,y)\hat{j}$

$F(0,0) = 0$ & $F(a,b)$ is tangent to $x^2+y^2 = a^2+b^2$, mag $= \sqrt{a^2+b^2}$

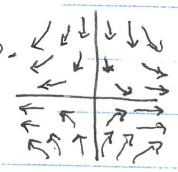
$$F(x,y) = y\hat{i} - x\hat{j}$$

32.



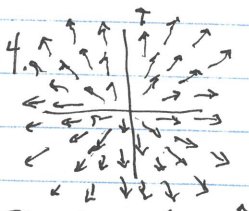
$G(x,y) = -y\hat{i} + x\hat{j}$

33.



$H(x,y) = x\hat{i} - y\hat{j}$

34.



$F(x,y) = x\hat{i} + y\hat{j}$

51. $F(x,y,z) = -\frac{1}{2}x\hat{i} - \frac{1}{2}y\hat{j} + \frac{1}{4}\hat{k}$

$r(t) = \cos t\,\hat{i} + \sin t\,\hat{j} + t\hat{k}$

from $(1,0,0)$ to $(-1,0,3\pi)$

$$\int_C F\cdot dr = \int_0^{3\pi} \langle -\tfrac{1}{2}\cos t, -\tfrac{1}{2}\sin t, \tfrac{1}{4}\rangle \cdot \langle -\sin t, \cos t, 1\rangle\, dt$$

$$= \int_0^{3\pi} \left(\tfrac{1}{2}\sin t\cos t - \tfrac{1}{2}\sin t\cos t + \tfrac{1}{4}\right) dt$$

$$= \int_0^{3\pi} \tfrac{1}{4}\, dt$$

$$= \tfrac{1}{4}t \Big|_0^{3\pi}$$

$$= \tfrac{3\pi}{4}$$

54. wire shaped as a circle of radius 2 centered at $(3,4)$ with $\rho(x,y) = y^2$

$r(t) = \langle 2\cos t + 3, 2\sin t + 4\rangle$

$$\int_0^{2\pi} (2\sin t + 4)^2 \sqrt{(-2\sin t)^2 + (2\cos t)^2}\, dt$$

$$= \int_0^{2\pi} (8\sin^2 t + 32\sin t + 32)\, dt$$

$$= \int_0^{2\pi} (4 - 4\cos 2t + 32\sin t + 32)\, dt$$

$$= (4t - 2\sin 2t - 32\cos t + 32t \Big|_0^{2\pi}$$

$$= (8\pi - 0 - 32 + 64\pi) - (0 - 0 - 32 + 0)$$

$$= 72\pi$$

57. $\int_C yz\,dx + xz\,dy + xy\,dz$
over $(1,1,1)$ to $(3,2,0)$

$r(t) = \langle 2t+1, t+1, -t-1 \rangle$
$r'(t) = \langle 2, 1, -1 \rangle$
$\int_0^1 [(t+1)(-t+1)2 + (2t+1)(-t+1)1 + (2t+1)(t+1)(-1)]\,dt$
$= \int_0^1 (-2t^2+2 - 2t^2+t+1 - 2t^2-3t-1)\,dt$
$= \int_0^1 (-6t^2-2t+2)\,dt$
$= (-2t^3-t^2+2t \Big|_0^1$
$= (-2-1+2) - 0$
$= -1$

94. wire bent into semicircle of radius a; linear mass density at point P is proportional to distance from line through endpoints. Find mass

$r(t) = \langle a\cos t, a\sin t \rangle$ $\quad \rho(x,y) = K\sqrt{x^2+y^2}$ $\quad$ for $K>0$
$r'(t) = \langle -a\sin t, a\cos t \rangle$
$\int_0^{2\pi} K\sqrt{(a\cos t)^2 + (a\sin t)^2}\,\sqrt{(-a\sin t)^2 + (a\cos t)^2}\,dt$

$= \int_0^{2\pi} K\sqrt{a^2(\cos^2 t + \sin^2 t)}\,\sqrt{a^2(\sin^2 t + \cos^2 t)}\,dt$

$= \int_0^{2\pi} Ka^2\,dt$
$= Ka^2 t \Big|_0^{2\pi}$
$= 2\pi Ka^2$